

Summary: Emerging Threats in AI

- **Core Objective:** The paper addresses the fragmented landscape of AI risk literature by synthesizing existing knowledge into a unified, **nine-domain analytical framework** of AI misuse, covering both technical mechanisms and socio-ethical impacts.
- **Key Finding:** There is a critical mismatch between the rapid advancement of AI capabilities and the defensive, governance, and regulatory mechanisms designed to mitigate them, giving adversaries a persistent advantage.

1. Global AI Incident Trends

Data aggregated from major repositories (such as the OECD AI Incidents Monitor and Stanford's AI Index Report) indicates an exponential surge in real-world AI failures and malicious misuses:

- **Exponential Growth:** Global AI incidents saw a **+104% growth** from 2023 (3,902 incidents) to 2024 (7,979 incidents).
- **Geographic Distribution:** In 2024, the United States recorded the highest volume of AI incidents, followed by India, China, Australia, and Germany.

2. The Nine Domains of AI Misuse

The paper consolidates fragmented risk repositories into nine primary categories:

Domain 1: Adversarial Threats

- **Evasion Attacks:** Subtle input perturbations (invisible to humans) designed to cross decision boundaries and cause high-confidence misclassifications in deployed models. *Example:* Tricking autonomous vehicle vision systems into misidentifying stop signs as yield signs.
- **Poisoning & Backdoor Attacks:** Injecting malicious samples into training data so the model develops a dormant vulnerability activated post-deployment by a specific attacker trigger (e.g., "BadNets").
- **Model Extraction:** Reconstructing a proprietary model's functionality locally by systematically querying its public API and analyzing output probability vectors.

Domain 2: Privacy Violations

- **Inference Attacks:** Membership inference (determining if a specific data point was in the training set) and model inversion (reconstructing sensitive features or original training data from model outputs).

- **Systemic Surveillance:** Mass biometric profiling and unauthorized facial recognition, enabling the tracking of individuals without consent.

Domain 3: Disinformation, Deception, and Propaganda

- **Synthetic Media:** The democratization of GAN- and diffusion-based deepfakes to create non-consensual intimate imagery or impersonate figures.
- **Automated Influence Operations:** Utilizing Large Language Models (LLMs) to generate persuasive disinformation at scale, accelerating the "liar's dividend" (the erosion of epistemic trust in authentic information).

Domain 4: Bias and Discrimination

- **Algorithmic Opacity:** Deep learning pipelines reflecting and reinforcing historical inequities due to skewed or unrepresentative training data.
- **Societal Impacts:** Quantified disparities in commercial gender classification, racially skewed recidivism tools, and biased healthcare resource allocation.

Domain 5: System Safety and Reliability Failures

- **Dual-Use Cybersecurity Threats:** AI being leveraged offensively to lower technical barriers to entry for constructing functional malware, automating high-precision phishing campaigns, and bypassing safety safeguards via adversarial prompting.
- **Operational Failures:** Autonomous vehicle perception errors and crashes resulting from a failure to navigate edge cases or unpredicted physical variables.

Domain 6: Socioeconomic Exploitation and Inequality

- **Market Manipulation:** Algorithmic pricing collusion, predatory microtargeting, and economic fraud targeting vulnerable populations.
- **Labor Disruption:** Accelerated structural job displacement across manufacturing and service sectors, driving economic disparities.

Domain 7: Environmental and Ecological Misuse

- **Carbon Footprint:** High energy consumption and carbon emissions stemming from large-scale model training and resource-intensive computations.
- **Ecological Harm:** Utilizing AI to optimize unsustainable industrial practices or actively circumvent environmental enforcement metrics.

Domain 8: Autonomy and Weaponization

- **Lethal Autonomous Weapons Systems (LAWS):** Delegating life-and-death decisions to machines, which lowers conflict thresholds and creates legal accountability voids.
- **Agentic AI Threats:** Emerging autonomous AI agents that can dynamically chain multi-step tasks (reconnaissance, privilege escalation, lateral movement) via tool/API use with minimal human oversight.

Domain 9: Human Interaction and Psychological Harm

- **Algorithmic Behavior Manipulation:** Utilizing persuasive, highly immersive digital environments to trigger dependency or addiction.
- **AI Companionship Harms:** Psychological risks arising from personal human-AI bonding, including emotional manipulation and dependency exploitation.

3. Empirical Case Studies Highlighted

The paper anchors its taxonomy in verified real-world incidents:

- **Clearview AI (Mass Surveillance):** Aggregated over 10 billion facial images from public platforms without user consent to build a database for law enforcement. *Legal outcome:* Multi-jurisdictional fines and regulatory restrictions under EU and Australian privacy frameworks.
- **Healthcare Resource Allocation Bias:** A commercial algorithm used health costs as a proxy for medical need. Because systemic barriers caused Black patients to incur lower medical costs for equal sickness levels, the model severely underestimated their healthcare needs. Correcting this proxy bias would have shifted the eligible Black patient care pool from 17.7% to 46.5%.
- **Voice-Cloning CEO Fraud:** In 2019, attackers trained commercial voice synthesis tools on public audio to spoof a CEO’s voice, successfully manipulating a subordinate into transferring \$243,000 to a fraudulent account.
- **Deepfake Non-Consensual Imagery:** Weaponized since 2017 to target women. Early frameworks lacked legislative provisions, but recent statutes in the UK and specific US states (California, Virginia) have criminalized this behavior.

4. Evaluation of Current Mitigation Strategies

The paper reviews a multi-layered defense landscape but notes steep operational trade-offs:

Approach Category	Core Techniques	Effectiveness	Major Limitations
Technical: Robustness	Adversarial training, Certified defenses	Low–Medium	High computational overhead; trade-offs with original model accuracy; fails against adaptive adversaries.

Technical: Detection	Forensic deepfake tools, GAN fingerprinting	Medium	Locked in an arms race; fails against novel or highly sophisticated generative models.
Technical: Privacy	Differential privacy, Federated learning	Medium–High	Significant implementation complexity; reduction in data utility.
Technical: Alignment	Constitutional AI, RLHF	Medium	Vulnerable to jailbreaking/adversarial prompting; can inherit human labeler biases.
Regulatory: Privacy	GDPR, CCPA	Medium–High	Severe cross-border enforcement challenges; cross-jurisdictional fragmented compliance burdens.
Regulatory: AI-Specific	EU AI Act, Sector rules	Unknown (Emerging)	Early implementation stage; compliance structures require deep technical expertise and auditing standardization.
Organizational	Ethics boards, Algorithmic impact audits	Low–Medium	Highly variable institutional commitment; lacks standardized independent authority.

5. Critical Governance and Regulatory Gaps

For the legal department, Section 6 highlights several core challenges:

- **Regulatory Arbitrage:** Because AI deployment is inherently borderless, conflicting regulations across global jurisdictions allow malicious actors or loose developers to move operations to weak legal environments.
- **The Inadequacy of Post-Hoc / Reactive Legal Measures:** Traditional tort law and privacy frameworks operate reactively. Once synthetic data or stolen biometric markers are leaked online, they cannot be retroactively "uncollected" or fully purged, creating permanent harm to victims.
- **Normative Conflict in Auditing:** While frameworks like the **EU AI Act** or **NIST AI RMF** mandate fairness and managed bias, mathematical definitions of fairness are often mutually incompatible. Striking a balance between accuracy, privacy, and fairness requires political and value-based choices rather than purely technical adjustments.